Lecture 24

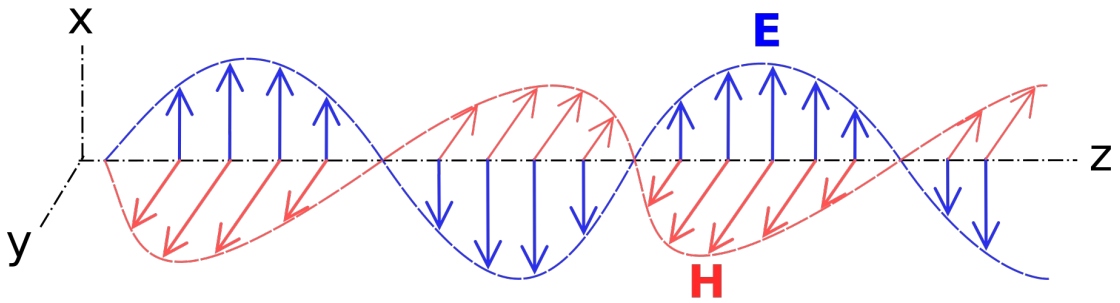
Polarization

Dr. Victoria Shao
yangshao@Illinois.edu

© Lynford Goddard, Brian Cunningham, and Victoria Shao
- Copying, publishing or distributing without express written permission is prohibited.

Definition of Polarization

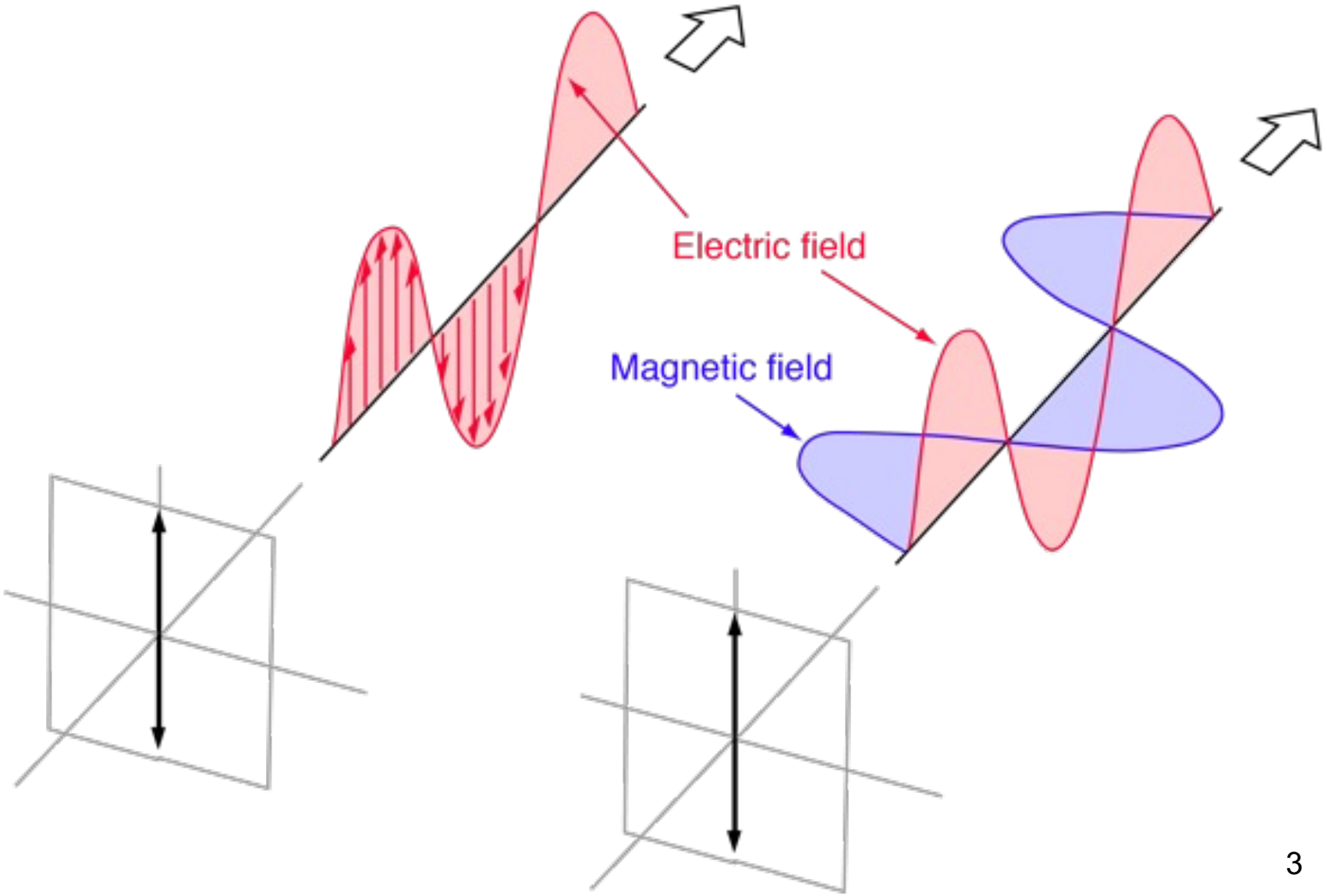
- Describes the "tip" of the time-varying **E** field vector for a particular point in space as it varies in time



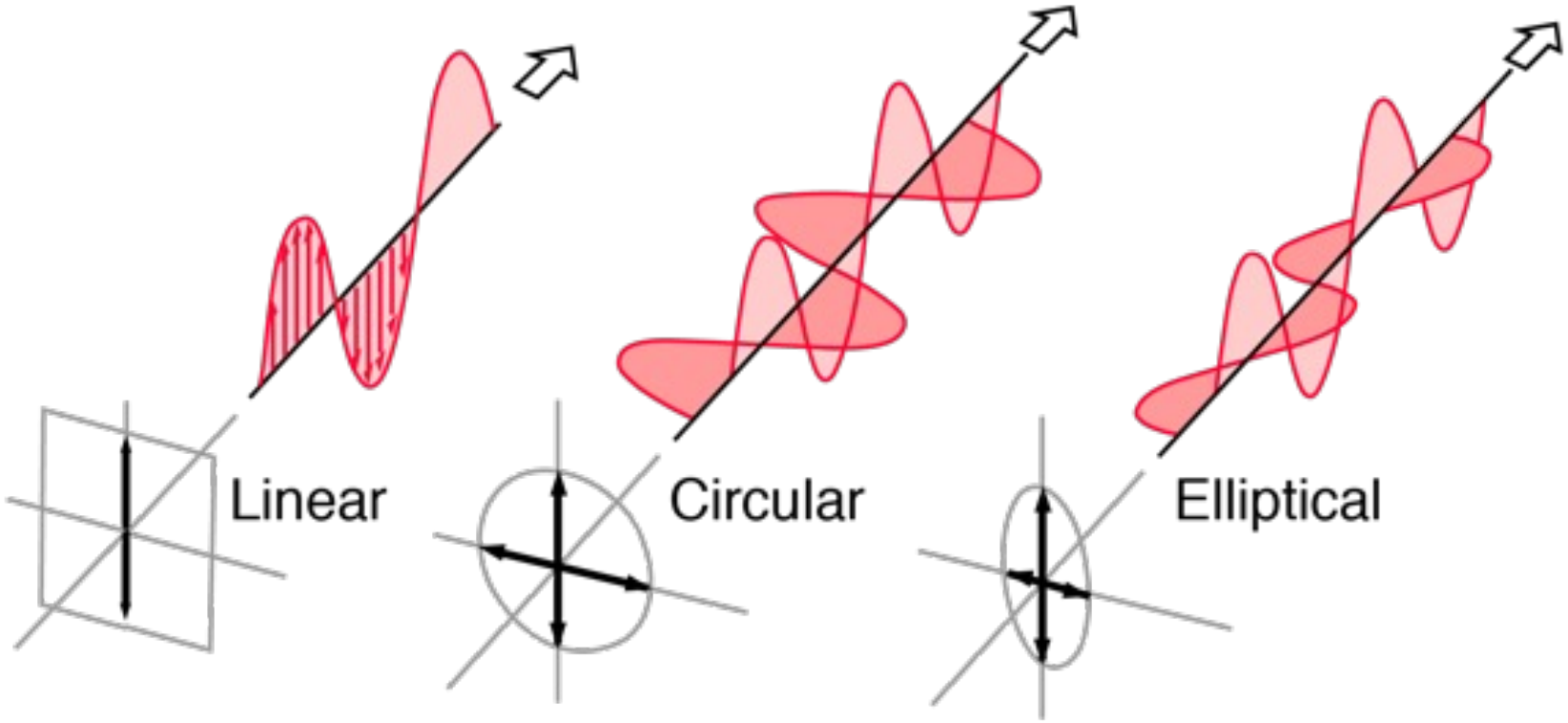
- $\vec{J} = -JS_0 cos(\omega t)\hat{a}_x$ example
  - Field with only one component is always linearly polarized

$$E_x(z,t) = \frac{\eta_0 J_{S0}}{2}\cos(\omega t \mp \beta z)$$

Linear Polarization



Classification of Polarization



How do we treat this mathematically?

Say we have two superimposed fields

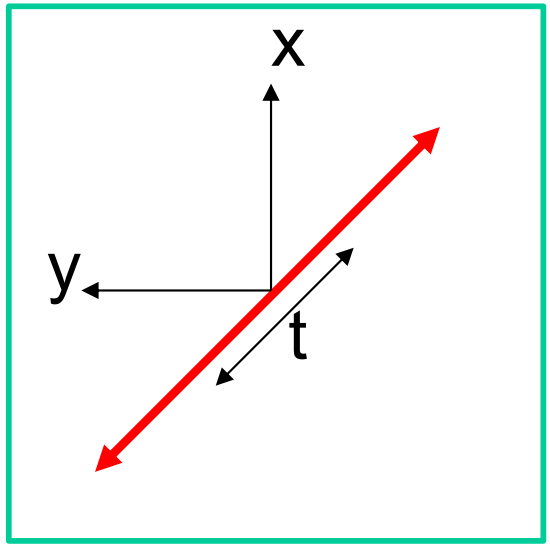
- Propagating in same direction
- Possibly out of phase
- Possibly oriented out of plane with each other

How can we tell if the combined wave will be:

- Linear
- Circular
- Elliptical

Linear Polarization

Plane in space perpendicular to propagation direction



The tip of the E-field vector traces out a straight line

DIRECTION: _____

MAGNITUDE: _____

Linear Polarization

Given two linearly polarized vector fields

$$\vec{E}_1 = E_1 \cos(\omega t + \phi)\hat{a}_x$$

$\mathbf{E}_1$ is horizontal linear polarization (red)

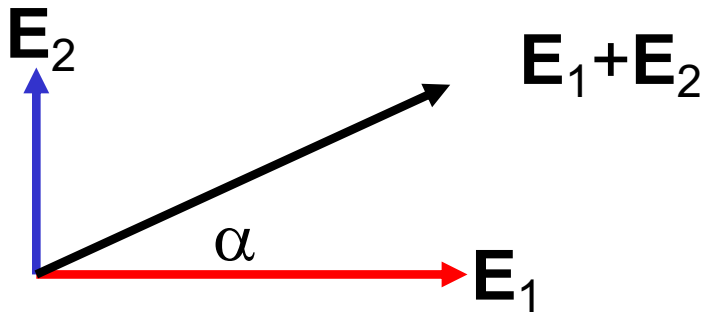
$$\vec{E}_2 = \pm E_2 \cos(\omega t + \phi)\hat{a}_y$$

$\mathbf{E}_2$ is vertical linear polarization (blue)

If the vectors are IN PHASE or 180° apart, $\mathbf{E}_1+\mathbf{E}_2$ will also be linear polarization

$$\vec{E}_1 + \vec{E}_2 = \cos(\omega t + \phi)(E_1\hat{a}_x \pm E_2\hat{a}_y)$$

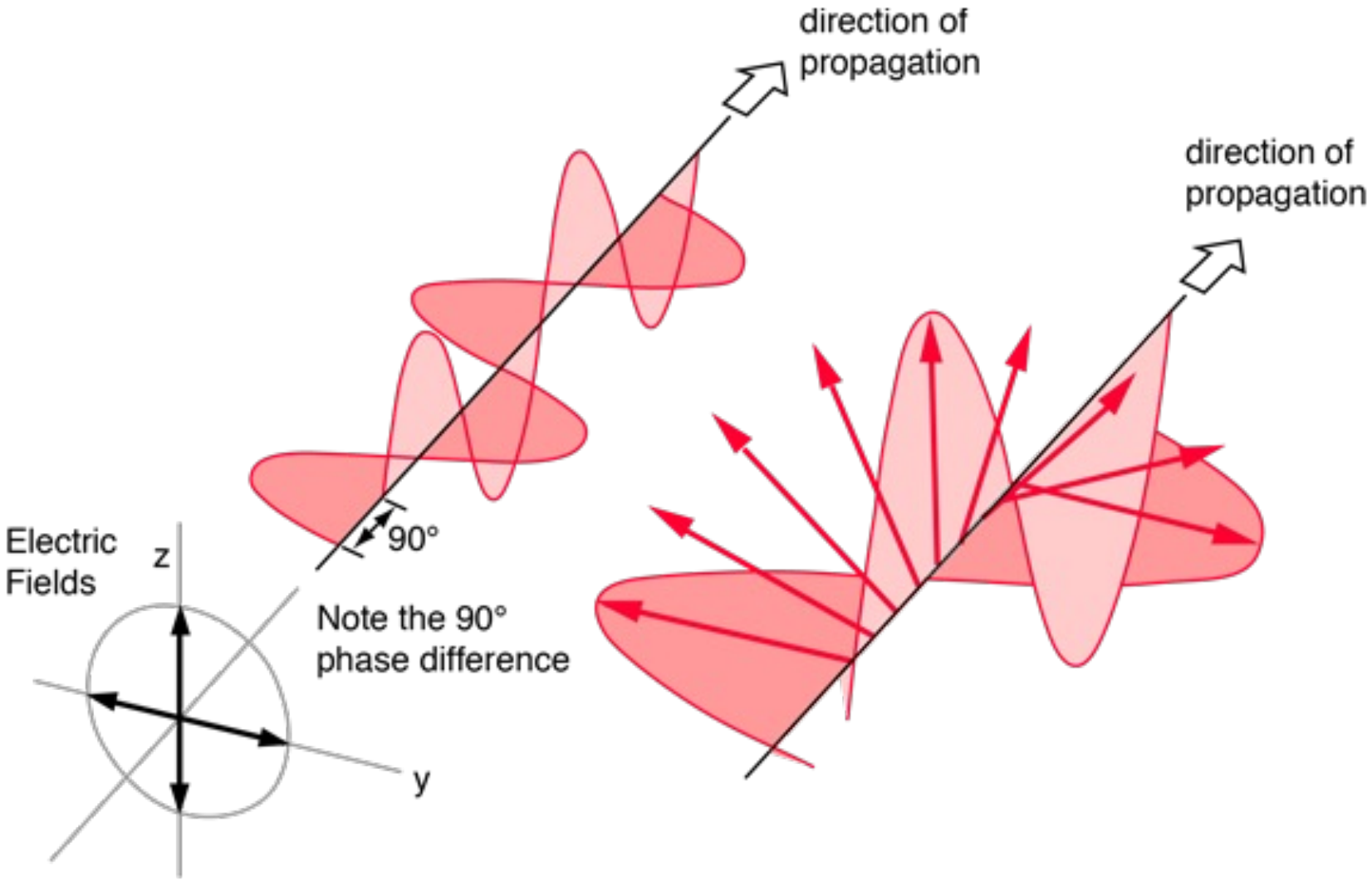
Magnitude changes Direction is constant



$$\alpha = \pm \tan^{-1}\left(\frac{E_2}{E_1}\right)$$

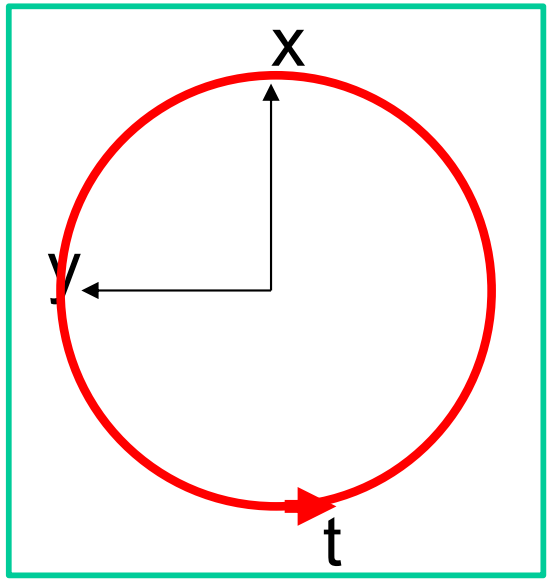
Polarization ANGLE depends on relative magnitudes of E_1 and E_2

Circular Polarization



Circular Polarization

Plane in space perpendicular to propagation direction



The tip of the E-field vector traces out a circle

DIRECTION: _____

MAGNITUDE: _____

Circular Polarization

Given two linearly polarized vector fields

$$\vec{E}_1 = E_0 \cos(\omega t + \phi)\hat{a}_x$$ **E**$_1$ is horizontal linear polarization

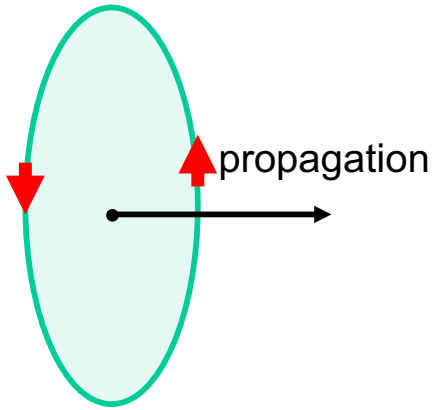
$$\vec{E}_2 = E_0 \sin(\omega t + \phi)\hat{a}_y$$ **E**$_2$ is vertical linear polarization

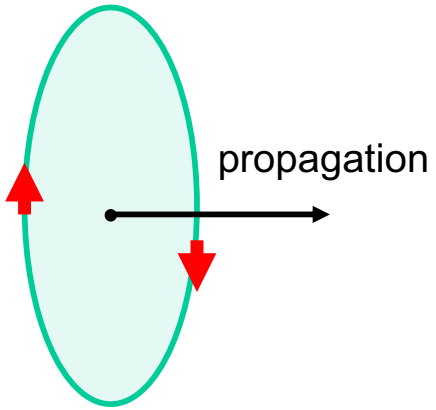
- ✓ The vectors have EQUAL MAGNITUDES
- ✓ The vectors are OUT OF PHASE by $\frac{\pi}{2}$
- ✓ The vectors are PERPENDICULAR

ALL THREE CONDITIONS MUST BE MET
for **E**$_1$+**E**$_2$ to be circular polarization

How to tell: Left or Right Handed Circular Polarization

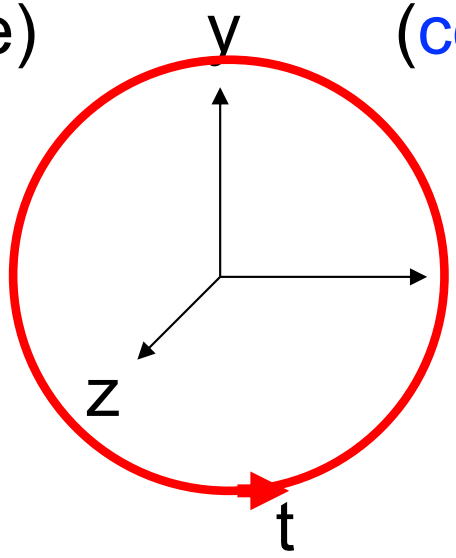
Method 1: Left/right thumb points in **propagation** direction





"Right-Hand" Polarized
(cw as seen by source)

"Left-Hand" Polarized
(ccw as seen by source)



☐ if moving in $-a_z$

☐ if moving in $+a_z$

How to tell: Left or Right Handed Circular Polarization

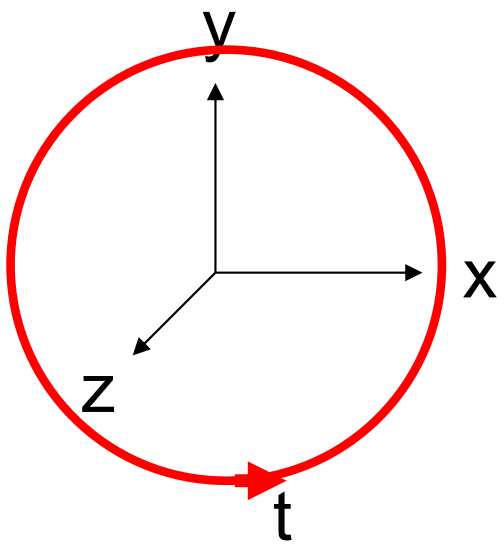
Method 2: It's right handed if
Ahead $\hat{E}$ x Behind $\hat{E}$ is in propagation direction

$$\vec{E}_1 = E_0 \cos(\omega t \pm \beta z)\hat{x} \qquad Ahead$$

$$\vec{E}_2 = E_0 \sin(\omega t \pm \beta z)\hat{y}$$

$$= E_0 \cos(\omega t \pm \beta z - \frac{\pi}{2})\hat{y} \quad Behind$$

It is behind by $\pi/2$ because ωt needs to be $\pi/2$ larger to be at the same part of the wave



$$\hat{E}_1 \times \hat{E}_2 = \hat{z}$$

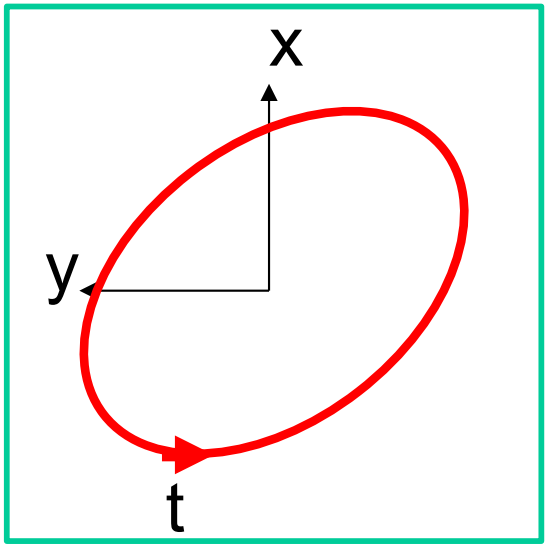
so ☐
if moving in $+a_z$

else ☐
if moving in $-a_z$

Elliptical Polarization

Most general: two combined linearly polarized waves will result in an elliptically polarized wave - if the conditions for linear or circular are not satisfied.

Plane in space perpendicular to propagation direction



The tip of the E-field vector traces out an ellipse

DIRECTION: _____

MAGNITUDE _____

Challenge Question: Forming Circular Polarization

For what value of ϕ, will the following fields add to produce **right**-handed circular polarization:

$$\vec{\boldsymbol{E}}_1 = E_0 \cos(2\pi\times 10^8 t + 2\pi z + \frac{\pi}{3})\hat{\boldsymbol{a}}_x$$

$$\vec{\boldsymbol{E}}_2 = E_0 \cos(2\pi\times 10^8 t + 2\pi z + \phi)\hat{\boldsymbol{a}}_y$$

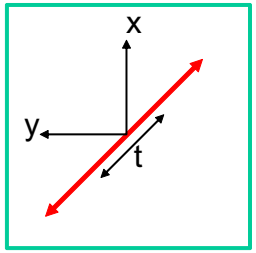
(a) $\pi/3$
(b) $-\pi/3$
(c) $\pi/6$
(d) $-\pi/6$
(e) $5\pi/6$

Writing linearly polarized as a sum of circular polarization

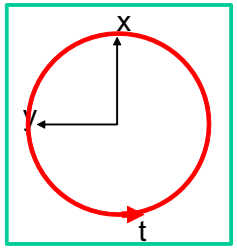
Rewrite linear polarized waves $\mathbf{E} = \mathrm{E}_0 \cos(\omega t + \beta z)\, \hat{a}_x$ as a sum of circular polarized fields.

Lecture 24 Summary

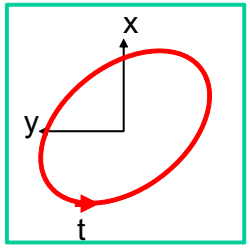
- Polarization describes the "tip" of **E**(t)
- Linear
  - Direction is _____
  - Magnitude _____



- Circular
  - Direction _____
  - Magnitude is _____



- Elliptical
  - Direction _____
  - Magnitude _____



- Next up: Wave reflection